Step 1: Explore Confidentiality

Objective: Understand how confidentiality is maintained and identify security violations when unauthorized access occurs.

1. Create a File Containing Sensitive Information

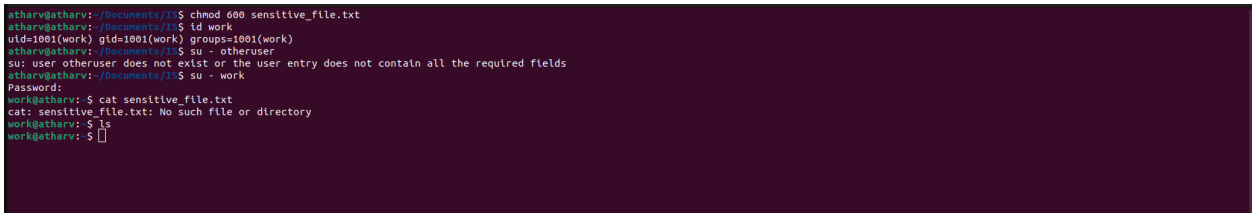
- Open a terminal or command prompt.
- Create a text file named sensitive_file.txt:
  Linux:
  echo "Confidential Data: Usernames and Passwords" > sensitive_file.txt
  Windows:
  Open Notepad, type Confidential Data: Usernames and Passwords, and save the file as sensitive_file.txt.

2. Restrict File Permissions

- Set file permissions so only the owner can access it:
  Linux:
  chmod 600 sensitive_file.txt
  Explanation: The 600 permission allows the owner to read/write the file but denies access to others.
  Windows:
  Right-click the file → Properties → Security → Edit permissions → Deny access for all users except the owner.

3. Simulate Unauthorized Access

- Switch to another user or simulate unauthorized access:
  Linux:
  su - otheruser
  cat sensitive_file.txt
  Windows:
  Switch user accounts or create a new user, then try opening the file.
- Observation:
  Expected error message: "Permission denied"
- Discussion:
  How this protects confidentiality.
  Real-world examples, such as protecting medical or financial records.



Step 2: Analyze Integrity

Objective: Understand how data integrity can be compromised and verify its integrity using
hashing.

1. Create or Access a Log File

   Use an existing log file or create a simulated one:
   Linux: sudo nano /var/log/syslog (requires root access).
   Windows: Open Event Viewer (eventvwr) or create a text file named 19logfile.txt.

2. Modify the Log File (Simulate Unauthorized Changes)

   Add or change log entries to simulate a security violation:
   Linux: Edit the file: sudo nano /var/log/syslog → Add a fake entry:
   Jan 1 12:00:00 UnauthorizedAccess: Admin login.
   Windows: Open logfile.txt in Notepad and add UnauthorizedAccess: Admin login.

3. Verify Integrity with Hashing

   Calculate the file's hash before and after modification:
   Linux: Use sha256sum logfile.txt and note the hash.
   Windows: Use PowerShell: Get-FileHash .\logfile.txt -Algorithm SHA256

Observe the hash difference.
Discussion: Discuss how unauthorized changes compromise data integrity. Relate this to tampering scenarios in real-world applications, such as altering financial records or audit logs.

```
  GNU nano 6.2                                                    /var/log/syslog *
Mar 22 18:00:49 atharv kernel: [    7.277789] rfkill: input handler disabled
Mar 22 18:00:49 atharv /usr/libexec/gdm-x-session[1794]: dbus-daemon[1794]: [session uid=128 pid=1794] Activating service name='org.gtk.vfs.Daemon' requested by ':1.19' (uid=128 pid=1900 comm="ibus-daemo
Mar 22 18:00:49 atharv gnome-shell[1821]: JS ERROR: TypeError: this._managerProxy is undefined#012_onGeoclueVanished@resource:///org/gnome/shell/ui/status/location.js:169:9
Mar 22 18:00:49 atharv /usr/libexec/gdm-x-session[1794]: dbus-daemon[1794]: [session uid=128 pid=1794] Activating service name='org.freedesktop.systemd1' requested by ':1.10' (uid=128 pid=1890 comm="/usr
Mar 22 18:00:49 atharv /usr/libexec/gdm-x-session[1794]: dbus-daemon[1794]: [session uid=128 pid=1794] Activated service 'org.freedesktop.systemd1' failed: Process org.freedesktop.systemd1 exited with st
Mar 22 18:00:49 atharv gsd-sharing[1890]: Failed to StopUnit service: GDBus.Error:org.freedesktop.DBus.Error.Spawn.ChildExited: Process org.freedesktop.systemd1 exited with status 1
Mar 22 18:00:49 atharv gsd-sharing[1890]: Failed to StopUnit service: GDBus.Error:org.freedesktop.DBus.Error.Spawn.ChildExited: Process org.freedesktop.systemd1 exited with status 1
Mar 22 18:00:49 atharv /usr/libexec/gdm-x-session[1794]: dbus-daemon[1794]: [session uid=128 pid=1794] Successfully activated service 'org.gtk.vfs.Daemon'
Mar 22 18:00:49 atharv /usr/libexec/gdm-x-session[1794]: dbus-daemon[1794]: [session uid=128 pid=1794] Activating service name='org.freedesktop.portal.IBus' requested by ':1.19' (uid=128 pid=1900 comm="i
Mar 22 18:00:49 atharv /usr/libexec/gdm-x-session[1794]: dbus-daemon[1794]: [session uid=128 pid=1794] Successfully activated service 'org.freedesktop.portal.IBus'
Mar 22 18:00:49 atharv dbus-daemon[737]: [system] Activating via systemd: service name='org.freedesktop.ColorManager' unit='colord.service' requested by ':1.51' (uid=128 pid=1898 comm="/usr/libexec/gsd-c
Mar 22 18:00:49 atharv systemd[1]: Starting Manage, Install and Generate Color Profiles...
Mar 22 18:00:49 atharv spice-vdagent[2000]: vdagent virtio channel /dev/virtio-ports/com.redhat.spice.0 does not exist, exiting
Mar 22 18:00:49 atharv gnome-session-binary[1795]: Entering running state
Mar 22 18:00:49 atharv NetworkManager[738]: <info>  [1742646649.9766] agent-manager: agent[9a8ec2cb97ddaaa9,:1.40/org.gnome.Shell.NetworkAgent/128]: agent registered
Mar 22 18:00:49 atharv gnome-shell[1821]: Error looking up permission: GDBus.Error:org.freedesktop.portal.Error.NotFound: No entry for geolocation
Mar 22 18:00:49 atharv xbrlapi.desktop[2003]: openConnection: connect: No such file or directory
Mar 22 18:00:49 atharv xbrlapi.desktop[2003]: cannot connect to braille devices daemon brltty at :0
Mar 22 18:00:49 atharv dbus-daemon[737]: [system] Successfully activated service 'org.freedesktop.ColorManager'
Mar 22 18:00:49 atharv systemd[1]: Started Manage, Install and Generate Color Profiles.
Mar 22 18:00:50 atharv NetworkManager[738]: <info>  [1742646650.0457] device (wlo1): supplicant interface state: scanning -> authenticating
Mar 22 18:00:50 atharv NetworkManager[738]: <info>  [1742646650.0457] device (p2p-dev-wlo1): supplicant management interface state: scanning -> authenticating
Mar 22 18:00:50 atharv kernel: [    7.406328] wlo1: send auth to 9c:d4:a6:f4:44:be (try 1/3)
Mar 22 18:00:50 atharv dbus-daemon[737]: [system] Successfully activated service 'org.freedesktop.GeoClue2'
Mar 22 18:00:50 atharv systemd[1]: Started Location Lookup Service.
Mar 22 18:00:50 atharv wpa_supplicant[769]: wlo1: Trying to associate with 9c:d4:a6:f4:44:be (SSID='HUN_5GJio' freq=5200 MHz)
Mar 22 18:00:50 atharv NetworkManager[738]: <info>  [1742646650.0487] device (wlo1): supplicant interface state: authenticating -> associating
Mar 22 18:00:50 atharv NetworkManager[738]: <info>  [1742646650.0487] device (p2p-dev-wlo1): supplicant management interface state: authenticating -> associating
Mar 22 18:00:50 atharv kernel: [    7.409319] wlo1: authenticated
Mar 22 18:00:50 atharv kernel: [    7.410302] wlo1: associate with 9c:d4:a6:f4:44:be (try 1/3)
Mar 22 18:00:50 atharv kernel: [    7.422422] wlo1: RX AssocResp from 9c:d4:a6:f4:44:be (capab=0x1111 status=0 aid=3)
Mar 22 18:00:50 atharv dbus-daemon[737]: [system] Successfully activated service 'org.freedesktop.UPower'
Mar 22 18:00:50 atharv systemd[1]: Started Daemon for power management.
Mar 22 18:00:50 atharv kernel: [    7.446652] wlo1: associated
Mar 22 18:00:50 atharv /usr/libexec/gdm-x-session[1794]: dbus-daemon[1794]: [session uid=128 pid=1794] Activating service name='org.gnome.ScreenSaver' requested by ':1.26' (uid=128 pid=1930 comm="/usr/lib
Mar 22 18:00:50 atharv gsd-media-keys[1916]: Failed to grab accelerator for keybinding settings:playback-repeat
Mar 22 18:00:50 atharv gsd-media-keys[1916]: Failed to grab accelerator for keybinding settings:rotate-video-lock
Mar 22 18:00:50 atharv gsd-media-keys[1916]: Failed to grab accelerator for keybinding settings:playback-random
Mar 22 18:00:50 atharv gsd-media-keys[1916]: Failed to grab accelerator for keybinding settings:hibernate
Mar 22 18:00:50 atharv /usr/libexec/gdm-x-session[1794]: dbus-daemon[1794]: [session uid=128 pid=1794] Successfully activated service 'org.gnome.ScreenSaver'
Mar 22 18:00:50 atharv /usr/libexec/gdm-x-session[1681]: (II) AMDGPU(0): EDID vendor "CMN", prod id 5432
Mar 22 18:00:50 atharv /usr/libexec/gdm-x-session[1681]: (II) AMDGPU(0): Printing DDC gathered Modelines:
Mar 22 18:00:50 atharv /usr/libexec/gdm-x-session[1681]: (II) AMDGPU(0): Modeline "1920x1080"x0.0  138.78  1920 1968 2000 2080  1080 1083 1088 1112 +hsync -vsync (66.7 kHz eP)
Mar 22 18:00:50 atharv /usr/libexec/gdm-x-session[1681]: (II) AMDGPU(0): Modeline "1920x1080"x0.0   92.52  1920 1968 2000 2080  1080 1083 1088 1112 +hsync -vsync (44.5 kHz e)
Mar 22 18:00:50 atharv /usr/libexec/gdm-x-session[1681]: (II) AMDGPU(0): EDID vendor "CMN", prod id 5432
Mar 22 18:00:50 atharv /usr/libexec/gdm-x-session[1681]: (II) AMDGPU(0): Printing DDC gathered Modelines:
Mar 22 18:00:50 atharv /usr/libexec/gdm-x-session[1681]: (II) AMDGPU(0): Modeline "1920x1080"x0.0  138.78  1920 1968 2000 2080  1080 1083 1088 1112 +hsync -vsync (66.7 kHz eP)
Mar 22 18:00:50 atharv /usr/libexec/gdm-x-session[1681]: (II) AMDGPU(0): Modeline "1920x1080"x0.0   92.52  1920 1968 2000 2080  1080 1083 1088 1112 +hsync -vsync (44.5 kHz e)
Mar 22 18:00:50 atharv /usr/libexec/gdm-x-session[1681]: (--) NVIDIA(GPU-0): DFP-0: disconnected
Jan 1 12:00:00 UnauthorizedAccess: Admin login.
Mar 22 18:00:50 atharv /usr/libexec/gdm-x-session[1681]: (--) NVIDIA(GPU-0): DFP-0: Internal TMDS
```

```
atharv@atharv:~/Documents/IS$ sudo nano /var/log/syslog
atharv@atharv:~/Documents/IS$ ls
sensitive_file.txt
atharv@atharv:~/Documents/IS$ sha256sum /var/log/syslog
a126b2b6740a4c8e7b385246ebcdb129a9d5bcde1571f533dfc12b1e71e44c00  /var/log/syslog
atharv@atharv:~/Documents/IS$
```

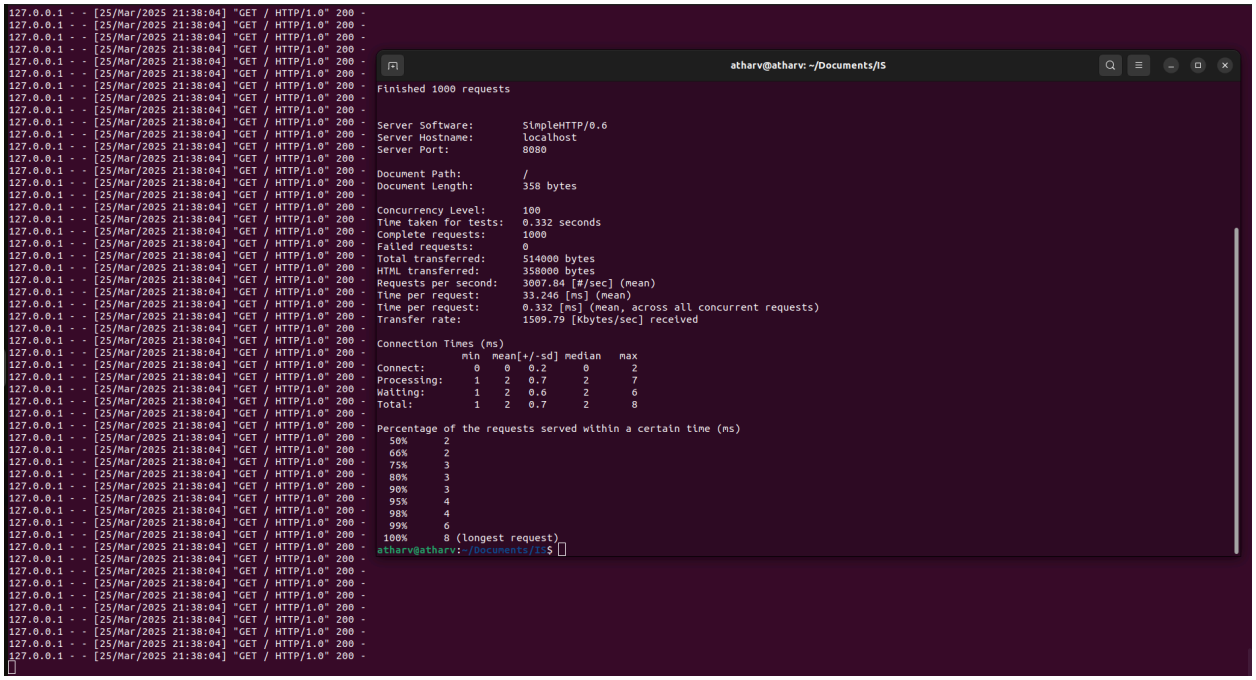
Step 3: Examine Availability
Objective: Explore how system availability is affected during an attack or resource overload.

1. Set Up a Simple Web Server (Optional)
Linux: Use Python to start a basic web server: python3 -m http.server 8080
Windows: Use IIS or WAMP/XAMPP to set up a local server.

2. Simulate Denial-of-Service (DoS) Attack
Use a tool to overload the server with requests:
Linux: Install and use ab (Apache Benchmark):
ab -n 1000 -c 100 http://localhost:8080/
-n: Total number of requests.
-c: Number of concurrent requests.
Windows: Use a custom PowerShell script or any load-testing tool like JMeter.

3. Monitor Server Behavior
Observe the server response time during the attack:
Linux: Check server logs or terminal output for delays or errors.
Windows: Use Task Manager or Resource Monitor to track CPU and network usage.
Note any timeouts or connection refusals.
204. Restore Normal Operations
Stop the attack and ensure normal availability:
Linux: Terminate the ab command or server process (Ctrl+C).
Windows: Stop the server or restart it via IIS Manager.
Verify that the server is responsive again.

5. Discussion
Relate this to real-world examples of denial-of-service attacks on popular websites or applications. Discuss strategies to mitigate such attacks, such as rate-limiting, firewalls, or load balancers.



Step 4: Document Observations

a. For each step, document:

- Actions performed.
- Observations (e.g., error messages, response times).
- Screenshots of commands and results.

b. Analyze how each violation impacts the overall security of the system.

Procedure/Program:

1. Discretionary Access Control (DAC)

a. Create Files and Directories:

Command

```
mkdir dac_demo
cd dac_demo
touch confidential.txt
```

What is happening?

mkdir dac_demo: Creates a directory named dac_demo.

touch confidential.txt: Creates an empty file named confidential.txt inside the directory.

b. Set Permissions Using chmod:

Command:

```
chmod 600 confidential.txt
```

The chmod 600 command changes the file's permissions:

Owner: Can read and write.

and Others: No permissions.

c. Change File Ownership Using chown:

Commands:

```
sudo adduser alice
sudo chown alice:alice confidential.txt
```

What is happening?

sudo adduser alice: Creates a new user named Alice.

sudo chown alice:alice confidential.txt: Transfers ownership of the file confidential.txt to Alice.

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d. Log in as Alice:

Command:

```
su alice
cat confidential.txt
```

What is happening?

su alice: Switches to the user Alice.

cat confidential.txt: Tries to read the file.

e. Results:

Alice, as the file owner, can access confidential.txt.

Other users (e.g., the default user) cannot read or write to the file because of the restrictive permissions (600).

```
atharv@atharv:~/Documents/IS$ mkdir dac_demo
atharv@atharv:~/Documents/IS$ cd dac_demo/
atharv@atharv:~/Documents/IS/dac_demo$ touch confidential.txt
atharv@atharv:~/Documents/IS/dac_demo$ chmod 600 confidential.txt
atharv@atharv:~/Documents/IS/dac_demo$ sudo adduser alice
[sudo] password for atharv:
sudo: a password is required
atharv@atharv:~/Documents/IS/dac_demo$ sudo adduser work
[sudo] password for atharv:
adduser: The user `work' already exists.
atharv@atharv:~/Documents/IS/dac_demo$ sudo adduser alice
Adding user `alice' ...
Adding new group `alice' (1002) ...
Adding new user `alice' (1002) with group `alice' ...
Creating home directory `/home/alice' ...
Copying files from `/etc/skel' ...
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
Changing the user information for alice
Enter the new value, or press ENTER for the default
        Full Name []: Atharv
        Room Number []: 1
        Work Phone []: 1
        Home Phone []: 1
        Other []: 1
Is the information correct? [Y/n] Y
atharv@atharv:~/Documents/IS/dac_demo$ sudo chown alice:alice confidential.txt
atharv@atharv:~/Documents/IS/dac_demo$ su alice
Password:
alice@atharv:/home/atharv/Documents/IS/dac_demo$ cat confidential.txt
alice@atharv:/home/atharv/Documents/IS/dac_demo$
```

2. Mandatory Access Control (MAC)

a. Enable SELinux:
Command : getenforce
What is happening?
Checks the status of SELinux (enforcing, permissive, or disabled).
If SELinux is not enabled:
sudo setenforce 1
This activates SELinux in "enforcing" mode.

b. Apply Security Context to a File:
Command: ls -Z confidential.txt
What is happening?
The ls -Z command displays the SELinux context of the file.
Example output: user_u:object_r:default_t:s0
confidential.txt
Change the context to allow access by specific processes:
sudo chcon -t httpd_sys_content_t confidential.txt
What is happening?
Changes the file's security context so that only the httpd
(web server) process can access it.

c. Test Policy Enforcement:
Try accessing the file through an unauthorized process.
Logs: sudo cat /var/log/audit/audit.log
What is happening?
SELinux denies access to unauthorized processes and logs
the event in audit.log.20

d. Results:
Access is granted or denied strictly based on SELinux policies,not ownership or file permissions.
Unauthorized processes will be denied access, even if they are run by the file owner.

```
atharv@atharv:~/Documents/IS$ getenforce
Disabled
atharv@atharv:~/Documents/IS$ sudo setenforce 1
setenforce: SELinux is disabled
atharv@atharv:~/Documents/IS$ ls
dac_demo  sensitive_file.txt
atharv@atharv:~/Documents/IS$ cd dac_demo/
atharv@atharv:~/Documents/IS/dac_demo$ ls
confidential.txt
atharv@atharv:~/Documents/IS/dac_demo$ ls -Z confidential.txt
? confidential.txt
atharv@atharv:~/Documents/IS/dac_demo$ sudo cat /var/log/audit/audit.log
cat: /var/log/audit/audit.log: No such file or directory
atharv@atharv:~/Documents/IS/dac_demo$
```

3. Role-Based Access Control (RBAC)

Create Roles (Groups):
Commands:
sudo groupadd managers
sudo usermod -aG managers alice
What is happening?
sudo groupadd managers: Creates a group called managers.
sudo usermod -aG managers alice: Adds Alice to the managers group.

Set Permissions on a File:
Commands:
touch manager_notes.txt
sudo chown :managers manager_notes.txt
sudo chmod 770 manager_notes.txt
What is happening?
touch manager_notes.txt: Creates a new file named manager_notes.txt.
o sudo chown :managers manager_notes.txt: Changes the file's group ownership to managers.
sudo chmod 770 manager_notes.txt: Sets permissions:
Owner and group: Read, write, and execute.
Others: No permissions.

Test Access:
As Alice (a member of the managers group):
su alice
cat manager_notes.txt
As another user (not in the managers group):
su <other_user>
cat manager_notes.txt
What is happening?
Alice can access the file because she belongs to the managers group.
Other users are denied access.

Results:

Access to manager_notes.txt is controlled by group membership (role), not individual ownership.

This approach simplifies access control in organizations where users' roles define their permissions.

```
atharv@atharv:~/Documents/IS/dac_demo$ sudo groupadd managers
atharv@atharv:~/Documents/IS/dac_demo$ sudo usermod -aG managers alice
atharv@atharv:~/Documents/IS/dac_demo$ touch manager_notes.txt
atharv@atharv:~/Documents/IS/dac_demo$ sudo chown :managers manager_notes.txt
atharv@atharv:~/Documents/IS/dac_demo$ sudo chmod 770 manager_notes.txt
atharv@atharv:~/Documents/IS/dac_demo$ su alice
Password:
alice@atharv:/home/atharv/Documents/IS/dac_demo$ cat manager_notes.txt
alice@atharv:/home/atharv/Documents/IS/dac_demo$ su work
Password:
work@atharv:/home/atharv/Documents/IS/dac_demo$ cat manager_notes.txt
cat: manager_notes.txt: Permission denied
work@atharv:/home/atharv/Documents/IS/dac_demo$
```

Procedure/Program:

Step 1: Setup File Access Policies (Confidentiality)

1. Create a Confidential File:
touch /home/student/secret.txt
echo "Confidential Data: Do not share!" > /home/student/secret.txt

2. Apply Strict File Permissions (DAC)
sudo chmod 600 "/home/student/secret.txt"
sudo chown root:root "/home/student/secret.txt"

3. Test Unauthorized Access
sudo -u nobody cat /home/student/secret.txt
Expected Outcome: Permission denied

Step 2: Enforce Integrity Policies Using Immutability

1. Enable File Immutability (Biba Model)
sudo chattr +i /home/student/secret.txt

2. Test Integrity Policy
sudo echo "Unauthorized Change" >> /home/student/secret.txt
Expected Outcome: Operation not permitted

3. Remove Immutability (Only When Required)
sudo chattr -i /home/student/secret.txt

Step 3: Implement Mandatory Access Control (MAC) using SELinux/AppArmor

1. Check SELinux/AppArmor Status
sestatus # For SELinux
sudo aa-status # For AppArmor

If disabled, enable SELinux:
sudo setenforce 1

2. Apply SELinux Labeling (MAC)
sudo semanage fcontext -a -t httpd_sys_content_t "/home/student/secret.txt"
sudo restorecon -v /home/student/secret.txt

Expected Outcome:
The file can only be accessed by HTTP services (if allowed).
Prevents unauthorized processes from reading/modifying it.

```
   /snap/firefox/5917/usr/lib/firefox/firefox (44889) snap.firefox.firefox
   /snap/firefox/5917/usr/lib/firefox/firefox (44990) snap.firefox.firefox
   /snap/firefox/5917/usr/lib/firefox/firefox (45030) snap.firefox.firefox
   /snap/firefox/5917/usr/lib/firefox/firefox (45145) snap.firefox.firefox
   /snap/firefox/5917/usr/lib/firefox/firefox (45168) snap.firefox.firefox
   /snap/firefox/5917/usr/lib/firefox/firefox (46433) snap.firefox.firefox
   /snap/firefox/5917/usr/lib/firefox/firefox (48650) snap.firefox.firefox
   /snap/snap-store/1216/usr/bin/snap-store (4702) snap.snap-store.ubuntu-software
   /snap/snapd-desktop-integration/253/usr/bin/snapd-desktop-integration (4231) snap.snapd-desktop-integration.snapd-desktop-integration
   /snap/snapd-desktop-integration/253/usr/bin/snapd-desktop-integration (4307) snap.snapd-desktop-integration.snapd-desktop-integration
0 processes are in complain mode.
0 processes are unconfined but have a profile defined.
0 processes are in mixed mode.
0 processes are in kill mode.
 thala7@thala7 :~/Documents/IS$ sudo setenforce 1
setenforce: SELinux is disabled
 thala7@thala7 :~/Documents/IS$ pwd
/home/atharv/Documents/IS
 thala7@thala7 :~/Documents/IS$ sudo semanage fcontext -a -t httpd_sys_content_t "/home/atharv/Documents/IS/secret.txt"
libsemanage.add_user: user sddm not in password file
 thala7@thala7 :~/Documents/IS$ sudo restorecon -v /home/atharv/Documents/IS/secret.txt
 thala7@thala7 :~/Documents/IS$
```

Step 4: Monitor Policy Violations Using Auditd

1. Install & Enable Auditd
sudo apt install auditd -y
sudo systemctl start auditd
sudo systemctl enable auditd

2. Add Audit Rule for Unauthorized Access
sudo auditctl -w /home/student/secret.txt -p war -k confidential_access

3. Check Audit Logs
sudo ausearch -k confidential_access --start today

```
 thala7@thala7 : $ sudo apt install auditd -y
[sudo] password for atharv:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  usb-modeswitch usb-modeswitch-data
Use 'sudo apt autoremove' to remove them.
Suggested packages:
  audispd-plugins
The following NEW packages will be installed:
  auditd
0 upgraded, 1 newly installed, 0 to remove and 5 not upgraded.
Need to get 212 kB of archives.
After this operation, 707 kB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 auditd amd64 1:3.0.7-1build1 [212 kB]
Fetched 212 kB in 1s (168 kB/s)
Selecting previously unselected package auditd.
(Reading database ... 335069 files and directories currently installed.)
Preparing to unpack .../auditd_1%3a3.0.7-1build1_amd64.deb ...
Unpacking auditd (1:3.0.7-1build1) ...
Setting up auditd (1:3.0.7-1build1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/auditd.service →/li
b/systemd/system/auditd.service.
Processing triggers for man-db (2.10.2-1) ...
 thala7@thala7 : $ sudo systemctl start auditd
 thala7@thala7 : $ sudo systemctl enable auditd
Synchronizing state of auditd.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable auditd
 thala7@thala7 : $
```

```
 thala7@thala7 : $ sudo auditctl -w /home/atharv/Documents/IS/secret.txt -p war -k confidential_access
 thala7@thala7 : $ sudo ausearch -k confidential_access --start today
----
time->Wed Mar 26 09:21:34 2025
type=PROCTITLE msg=audit(1742961094.959:490): proctitle=617564697463746C002D77002F686F6D652F6174686172762F446F63756D656E74732F49532F73656372265742E747874002D7000776172002D6B00636F6E666964656E7469616C5F6163
63657373
type=SYSCALL msg=audit(1742961094.959:490): arch=c000003e syscall=44 success=yes exit=1112 a0=4 a1=7fff7c94f1b0 a2=458 a3=0 items=0 ppid=50568 pid=50569 auid=1000 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0
sgid=0 fsgid=0 tty=pts2 ses=6 comm="auditctl" exe="/usr/sbin/auditctl" subj=unconfined key=(null)
type=CONFIG_CHANGE msg=audit(1742961094.959:490): auid=1000 ses=6 subj=unconfined op=add_rule key="confidential_access" list=4 res=1
 thala7@thala7 : $
```

Procedure/Program:

Part 1: Identity-Based Access Using PKI

Step 1: Install Required Software

1. Update and Upgrade Ubuntu

sudo apt update && sudo apt upgrade -y

2. Install OpenSSL for Certificate Management

sudo apt install openssl -y

openssl version # Verify installation

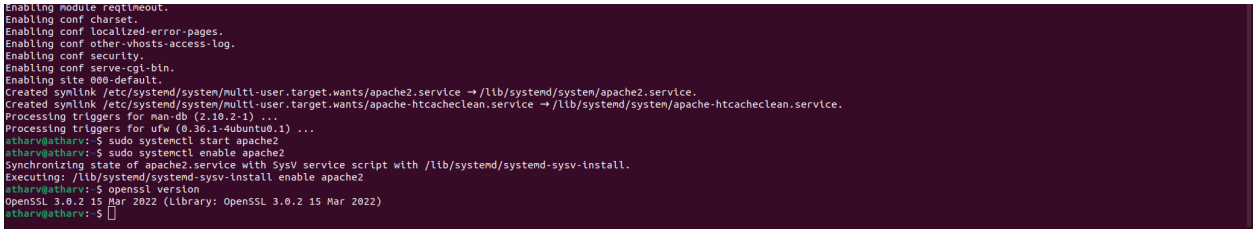
3. Install Apache Web Server

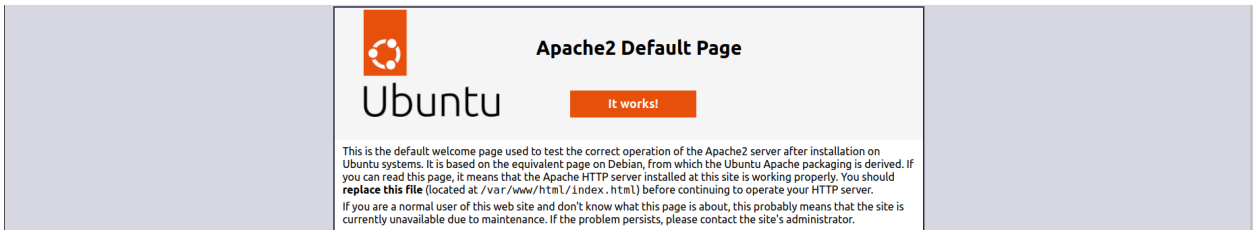
sudo apt install apache2 -y

sudo systemctl start apache2

sudo systemctl enable apache2

Test Apache: Open http://localhost in your browser. You should see the Apache default page.





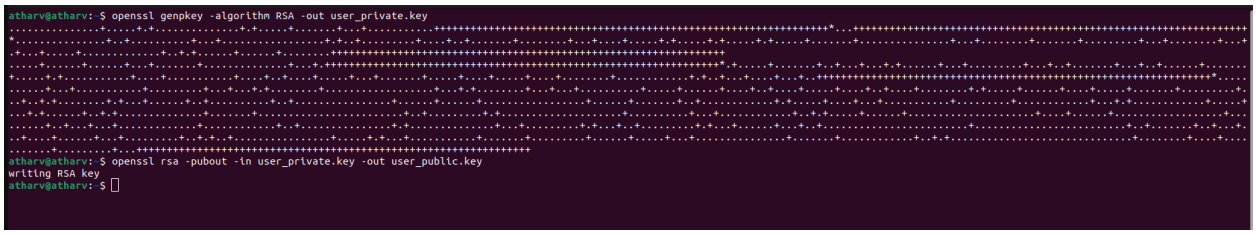
Step 2: Generate Public/Private Key Pair

1. Generate the User's Private Key

openssl genpkey -algorithm RSA -out user_private.key

2. Extract the Public Key

openssl rsa -pubout -in user_private.key -out user_public.key



Step 3: Create & Sign a User Certificate

1. Create a Certificate Signing Request (CSR)

openssl req -new -key user_private.key -out user.csr

When prompted, enter details like:

Country Name: US
Organization Name: SecureLab
Common Name: User1

2. Create a Self-Signed Certificate

openssl x509 -req -in user.csr -signkey user_private.key -out user.crt

```
atharv@atharv: $ openssl req -new -key user_private.key -out user.csr
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:US
State or Province Name (full name) [Some-State]:Calfornia
Locality Name (eg, city) []:LA
Organization Name (eg, company) [Internet Widgits Pty Ltd]:SecureLab
Organizational Unit Name (eg, section) []:.
Common Name (e.g. server FQDN or YOUR name) []:User1
Email Address []:xyz@gmail.com

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:123456789
An optional company name []:xyz
atharv@atharv: $ openssl x509 -req -in user.csr -signkey user_private.key -out user.crt
Certificate request self-signature ok
subject=C = US, ST = Calfornia, L = LA, O = SecureLab, CN = User1, emailAddress = xyz@gmail.com
atharv@atharv: $
```

Step 4: Set Up a Certificate Authority (CA)

1. Create the CA Private Key

openssl genpkey -algorithm RSA -out ca_private.key

2. Create the CA Certificate

openssl req -key ca_private.key -new -x509 -out ca_certificate.crt

3. Sign the User's Certificate Using the CA

openssl x509 -req -in user.csr -CA ca_certificate.crt -CAkey ca_private.key -CAcreateserial -out user_signed.crt

Now, user_signed.crt is issued and signed by the CA.

```
atharv@atharv:~$ openssl genpkey -algorithm RSA -out ca_private.key
.+.+...+.........+...+..+.........+.....+......+....+...++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++*....++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++*....+......
...+........+..+...+.........+......+.....+......+..+..+.+................+............+.+............+..+.......+.....+.........+.+...+...+........+.....................+...........+.....+......+....+..
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+...++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
atharv@atharv:~$ openssl req -key ca_private.key -new -x509 -out ca_certificate.crt
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:US
State or Province Name (full name) [Some-State]:california
Locality Name (eg, city) []:LA
Organization Name (eg, company) [Internet Widgits Pty Ltd]:SecureLab
Organizational Unit Name (eg, section) []:.
Common Name (e.g. server FQDN or YOUR name) []:User1
Email Address []:xyz@gmail.com
atharv@atharv:~$ openssl x509 -req -in user.csr -CA ca_certificate.crt -CAkey ca_private.key
-CAcreateseserial -out user_signed.crt
Certificate request self-signature ok
subject=C = US, ST = Calfornia, L = LA, O = SecureLab, CN = User1, emailAddress = xyz@gmail.com
-----BEGIN CERTIFICATE-----
MIIDaDCCAlACFB1RyNQVnJwStVnybcdDCZZOpn/9MA0GCSqGSIb3DQEBCwUAMHEx
CzAJBgNVBAYTAlVTMRMwEQYDVQQIDApjYWxpZm9ybmlhMQswCQYDVQQHDAJMQTES
MBAGA1UECgwJU2VjdXJlTGFiMQ4wDAYDVQQDDAVVc2VyMTEcMBoGCSqGSIb3DQEJ
ARYNeHl6QGdtYWlsLmNvbTAeFw0yNTAzMjUxNTI0MDBaFw0yNTA0MjQxNTI0MDBa
MHAxCzAJBgNVBAYTAlVTMRIwEAYDVQQIDAlDYWxmb3JuaWExCzAJBgNVBAcMAkxB
MRIwEAYDVQQKDAlTZWN1cmVMYWIxDjAMBgNVBAMMBVVzZXIxMRwwGgYJKoZIhvcN
AQkBFg14eXpAZ21haWwuY29tMIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKC
AQEAw7pMdiuPHiwFChs+EGanittmpi3FNfsY6TdvzT3FdbTVguK4s7CvHzqyFt+X
9B/k/DyZTrmap9Ws6NUpVPHf1UQ0rPqkXzs3vqkIfjxEPIcdsrW0TOIK6QC/fsRE
uGgTlG8QHkx3mo93A+wQue5sOUdVrzovDo8y8bwKuyEm3tOwlkssQ/lD5pyHzsOF
gcSYRTTcJedo1WY+tooDxWdTTWwU5+qlT23JqX3JfqhmXbrbFBAf/DEQ+xY+NHwtl
6BUANMKUilIRhASAFsF44Ba7RUTUYHKazMUSyNIov4Clk1b8gO6ckHU/cpI6kbnK
w2xVinf39wOnWGNdicU5kLG7wQIDAQABMA0GCSqGSIb3DQEBCwUAA4IBAQCl0FDq
rsUTd7NfR3IvhG7HWIXNebAv5oR5QojNmD/nOhVpBPuB3OvPiHplxW8nzBnCB7wg
ljVNOqZNAosy9X94/F1a7C3jq9rOdor8A052AfMgqMxlJn7qI1z7A5Vq6N9wHI4l
WlMrVnct3XI4pb9zkAU04w7E49hQtVYl6rvvUFA59z2lcpVRs6nnesgrmrLh3++j
EGxz1UXfVfIs2jKmITkXO5Ce/WozSajuC0lleccQUd3HBvfu1FCF2aj9NRjJeCf0
wz1ILAnOPKjKKXZY3Cp4Y8AAuI10BTUPTaF1wsOekJlmEhMerpNh3SG6yTiohr/B
dVRUA/MyCWGRwr0k
-----END CERTIFICATE-----
-CAcreateserial: command not found
atharv@atharv:~$
```

Step 5: Configure Apache for Identity-Based Access

1. Enable SSL Module

sudo a2enmod ssl

2. Configure SSL Certificate in Apache

Edit the Apache SSL configuration:

sudo nano /etc/apache2/sites-available/default-ssl.conf

Replace <VirtualHost *:443> section with:

<VirtualHost *:443>
DocumentRoot /var/www/html
SSLEngine on
SSLCertificateFile /etc/ssl/certs/user_signed.crt
SSLCertificateKeyFile /etc/ssl/private/user_private.key
SSLCACertificateFile /etc/ssl/certs/ca_certificate.crt
SSLVerifyClient require
SSLVerifyDepth 1
</VirtualHost>

3. Restart Apache

sudo systemctl restart apache2

```
atharv@atharv:~$ sudo a2enmod ssl
Considering dependency setenvif for ssl:
Module setenvif already enabled
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create self-signed certificates.
To activate the new configuration, you need to run:
  systemctl restart apache2
atharv@atharv:~$ sudo nano /etc/apache2/sites-available/default-ssl.conf
atharv@atharv:~$ sudo systemctl restart apache2
atharv@atharv:~$
```

Step 6: Create a Protected Directory

1. Create a Secure Folder

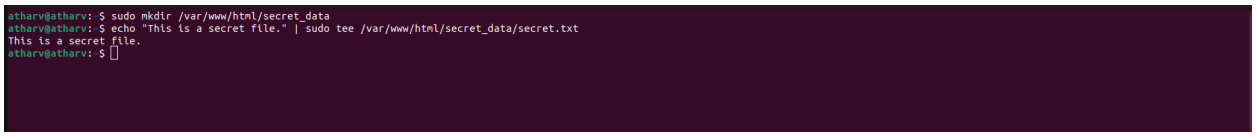
sudo mkdir /var/www/html/secret_data

echo "This is a secret file." | sudo tee /var/www/html/secret_data/secret.txt

2. Restrict Access to the Directory

Apache will require valid certificates to access /secret_data/.

Apache is now configured for IBAC!



Step 7: Simulate the Confinement Problem

1. Create Python Script

nano access_control.py

2. Add the Following Code

```
# Function to check access based on clearance levels
def check_access(user_clearance, resource_clearance):
if user_clearance >= resource_clearance:
return "Access Granted"
Else:
return "Access Denied"
# Clearance Levels
clearance_levels = {
1: "Top Secret",
2: "Secret",
3: "Confidential",
4: "Unclassified"
}
# Get user input
print("Available Clearance Levels:")
for level, name in clearance_levels.items():
print(f"{level}: {name}")
user_clearance = int(input("Enter your clearance level (1-4): "))
resource_clearance = int(input("Enter the clearance level of the resource (1-4):
"))
# Check Access
access_result = check_access(user_clearance, resource_clearance)
# Display Result
print(f"Your Clearance Level: {clearance_levels[user_clearance]}")
print(f"Resource Clearance Level: {clearance_levels[resource_clearance]}")
20print(access_result)
```

3. Code Run the Script
python3 access_control.py

```
  GNU nano 6.2                                                    access_control.py *
# Function to check access based on clearance levels
def check_access(user_clearance, resource_clearance):
    if user_clearance >= resource_clearance:
        return "Access Granted"
    else:
        return "Access Denied"

# Clearance Levels
clearance_levels = {
    1: "Top Secret",
    2: "Secret",
    3: "Confidential",
    4: "Unclassified"
}

# Get user input
print("Available Clearance Levels:")
for level, name in clearance_levels.items():
    print(f"{level}: {name}")

user_clearance = int(input("Enter your clearance level (1-4): "))
resource_clearance = int(input("Enter the clearance level of the resource (1-4): "))

# Check Access
access_result = check_access(user_clearance, resource_clearance)

# Display Result
print(f"Your Clearance Level: {clearance_levels[user_clearance]}")
print(f"Resource Clearance Level: {clearance_levels[resource_clearance]}")
print(access_result)

File Name to Write: access_control.py
```

```
atharv@atharv:~$ nano access_control.py
atharv@atharv:~$ python3 access_control.py
Available Clearance Levels:
1: Top Secret
2: Secret
3: Confidential
4: Unclassified
Enter your clearance level (1-4): 1
Enter the clearance level of the resource (1-4): 1
Your Clearance Level: Top Secret
Resource Clearance Level: Top Secret
Access Granted
atharv@atharv:~$ python3 access_control.py
Available Clearance Levels:
1: Top Secret
2: Secret
3: Confidential
4: Unclassified
Enter your clearance level (1-4): 2
Enter the clearance level of the resource (1-4): 3
Your Clearance Level: Secret
Resource Clearance Level: Confidential
Access Denied
atharv@atharv:~$ python3 access_control.py
Available Clearance Levels:
1: Top Secret
2: Secret
3: Confidential
4: Unclassified
Enter your clearance level (1-4): 4
Enter the clearance level of the resource (1-4): 1
Your Clearance Level: Unclassified
Resource Clearance Level: Top Secret
Access Granted
atharv@atharv:~$
```

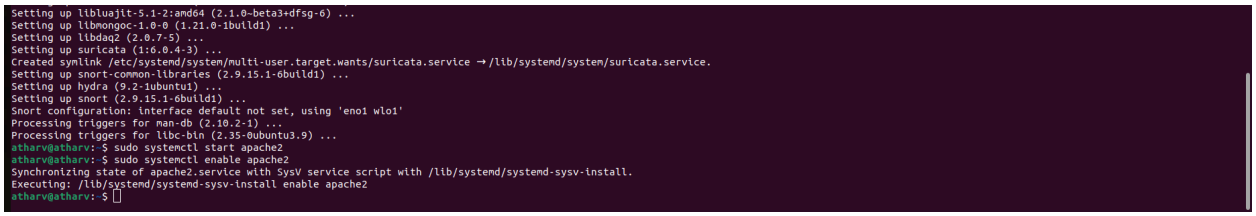
Procedure/Program:

Step 1: Update Ubuntu and Install Required Packages

1. Update the package repository:
sudo apt update && sudo apt upgrade -y

2. Install security tools:
sudo apt install snort suricata nmap nikto hydra apache2 -y

3. Enable Apache web server:
sudo systemctl start apache2
sudo systemctl enable apache2



Part 1: Configuring Intrusion Detection Systems (IDS)

Step 2: Configure Snort IDS

1. Verify Snort Installation:
snort -V

2. Edit Snort Configuration File:
sudo nano /etc/snort/snort.conf
Set the HOME_NET to match your network:
ipvar HOME_NET 10.10.10.0/24

preprocessor sfportscan: proto { all } scan_type { all } sense_level { low }
output alert_fast: /var/log/snort/alert

3. Run Snort in IDS Mode:
sudo snort -A console -c /etc/snort/snort.conf -i eth0
(eth0: interface name, check ifconfig for the interface name and IP address)

```
4057 Snort rules read
    3383 detection rules
    0 decoder rules
    0 preprocessor rules
3383 Option Chains linked into 949 Chain Headers
+++++++++++++++++++++++++++++++++++++++++++++++++++++++++++

+-------------------[Rule Port Counts]--------------------------------------
|             tcp     udp    icmp      ip
|     src     151      18       0       0
|     dst    3306     126       0       0
|     any     383      48      52      22
|      nc      27       8      15      20
|     s+d      12       5       0       0
+----------------------------------------------------------------------------

+-----------------------[detection-filter-config]-----------------------------
| memory-cap : 1048576 bytes
+-----------------------[detection-filter-rules]------------------------------
| none
-----------------------------------------------------------------------------

+-----------------------[rate-filter-config]----------------------------------
| memory-cap : 1048576 bytes
+-----------------------[rate-filter-rules]-----------------------------------
| none
-----------------------------------------------------------------------------

+-----------------------[event-filter-config]---------------------------------
| memory-cap : 1048576 bytes
+-----------------------[event-filter-global]---------------------------------
| none
+-----------------------[event-filter-local]----------------------------------
| gen-id=1      sig-id=1991       type=Limit     tracking=src count=1   seconds=60
| gen-id=1      sig-id=2923       type=Threshold tracking=dst count=10  seconds=60
| gen-id=1      sig-id=2495       type=Both      tracking=dst count=20  seconds=60
| gen-id=1      sig-id=2924       type=Threshold tracking=dst count=10  seconds=60
| gen-id=1      sig-id=3152       type=Threshold tracking=src count=5   seconds=2
| gen-id=1      sig-id=2496       type=Both      tracking=dst count=20  seconds=60
| gen-id=1      sig-id=2275       type=Threshold tracking=dst count=5   seconds=60
| gen-id=1      sig-id=2494       type=Both      tracking=dst count=20  seconds=60
| gen-id=1      sig-id=2523       type=Both      tracking=dst count=10  seconds=10
| gen-id=1      sig-id=3273       type=Threshold tracking=src count=5   seconds=2
+-----------------------[suppression]----------------------------------------
| none
-----------------------------------------------------------------------------
Rule application order: pass->drop->sdrop->reject->alert->log
Verifying Preprocessor Configurations!
WARNING: flowbits key 'smb.tree.create.llsrpc' is set but not ever checked.
WARNING: flowbits key 'ms_sql_seen_dns' is checked but not ever set.
33 out of 1024 flowbits in use.

[ Port Based Pattern Matching Memory ]
+- [ Aho-Corasick Summary ] -----------------------------------
| Storage Format    : Full-Q
```

```
| memory-cap : 1048576 bytes
+-----------------------[event-filter-global]---------------------------------
| none
+-----------------------[event-filter-local]----------------------------------
| gen-id=1      sig-id=1991       type=Limit     tracking=src count=1   seconds=60
| gen-id=1      sig-id=2923       type=Threshold tracking=dst count=10  seconds=60
| gen-id=1      sig-id=2495       type=Both      tracking=dst count=20  seconds=60
| gen-id=1      sig-id=2924       type=Threshold tracking=dst count=10  seconds=60
| gen-id=1      sig-id=3152       type=Threshold tracking=src count=5   seconds=2
| gen-id=1      sig-id=2496       type=Both      tracking=dst count=20  seconds=60
| gen-id=1      sig-id=2275       type=Threshold tracking=dst count=5   seconds=60
| gen-id=1      sig-id=2494       type=Both      tracking=dst count=20  seconds=60
| gen-id=1      sig-id=2523       type=Both      tracking=dst count=10  seconds=10
| gen-id=1      sig-id=3273       type=Threshold tracking=src count=5   seconds=2
+-----------------------[suppression]----------------------------------------
| none
-----------------------------------------------------------------------------
Rule application order: pass->drop->sdrop->reject->alert->log
Verifying Preprocessor Configurations!
WARNING: flowbits key 'smb.tree.create.llsrpc' is set but not ever checked.
WARNING: flowbits key 'ms_sql_seen_dns' is checked but not ever set.
33 out of 1024 flowbits in use.

[ Port Based Pattern Matching Memory ]
+- [ Aho-Corasick Summary ] -----------------------------------
| Storage Format    : Full-Q
| Finite Automaton  : DFA
| Alphabet Size     : 256 Chars
| Sizeof State      : Variable (1,2,4 bytes)
| Instances         : 215
|     1 byte states : 204
|     2 byte states : 11
|     4 byte states : 0
| Characters        : 64755
| States            : 31951
| Transitions       : 863868
| State Density     : 10.6%
| Patterns          : 5041
| Match States      : 3836
| Memory (MB)       : 16.90
|   Patterns        : 0.51
|   Match Lists     : 1.01
|   DFA
|     1 byte states : 1.02
|     2 byte states : 13.96
|     4 byte states : 0.00
+-[ Number of patterns truncated to 20 bytes: 1038 ]----------------------
pcap DAQ configured to passive.
Acquiring network traffic from "eth0".
Reload thread starting...
Reload thread started, thread 0x70a8f0800640 (54393)
ERROR: Can't start DAQ (-1) - SIOCGIFHWADDR: No such device!
Fatal Error, Quitting..
atharv@atharv: $
```

Part 2: Performing Vulnerability Analysis

Step 3: Scan for Open Ports Using Nmap

1. Identify the Target IP Address:

   ip a

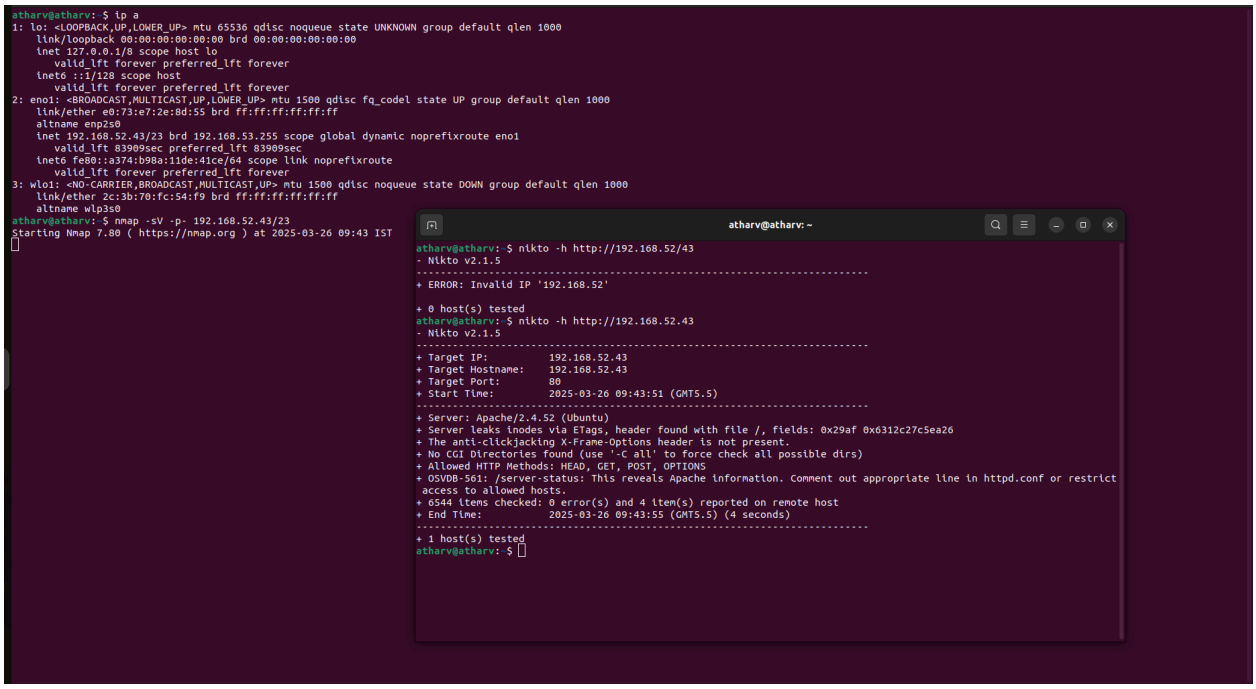
2. Run Nmap Scan:

   nmap -sV -p- 10.10.10.6 (Check for your system IP address here)

Step 4: Web Vulnerability Scan Using Nikto

1. Run Nikto Scan:

nikto -h http://10.10.10.6 (Check for your system IP address here)



Part 3: Performing Vulnerability Analysis

Step 5: Simulate Port Scanning Attack

1. Run an aggressive Nmap scan:
   nmap -A 10.10.10.6 (Check for your system IP address here)

2. Check Snort Logs:
   sudo cat /var/log/snort/alert

```
atharv@atharv:~$ ps aux | grep snort
snort      53020  0.0  0.9 200652 153920 ?       Ssl  09:26   0:00 /usr/sbin/snort -m 027 -D -d -l /var/log/snort -u snort -g snort --pid-path /run/snort/ -c /etc/snort/snort.conf -S "HOME_NET=[192.168.0.
0/16]" -i eno1
snort      53035  0.0  0.5 131616 84632 ?        Ssl  09:26   0:00 /usr/sbin/snort -m 027 -D -d -l /var/log/snort -u snort -g snort --pid-path /run/snort/ -c /etc/snort/snort.conf -S "HOME_NET=[192.168.0.
0/16]" -i wlo1
atharv     55821  0.0  0.0   9532  2560 pts/0    S+   09:49   0:00 grep --color=auto snort
atharv@atharv:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eno1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether e0:73:e7:2e:8d:55 brd ff:ff:ff:ff:ff:ff
    altname enp2s0
    inet 192.168.52.43/23 brd 192.168.53.255 scope global dynamic noprefixroute eno1
       valid_lft 83437sec preferred_lft 83437sec
    inet6 fe80::a374:b98a:11de:41ce/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: wlo1: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default qlen 1000
    link/ether 2c:3b:70:fc:54:f9 brd ff:ff:ff:ff:ff:ff
    altname wlp3s0
atharv@atharv:~$ nmap -Pn -A 192.168.52.43
Starting Nmap 7.80 ( https://nmap.org ) at 2025-03-26 09:50 IST
Stats: 0:00:56 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 91.67% done; ETC: 09:51 (0:00:01 remaining)
Stats: 0:01:12 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 91.67% done; ETC: 09:51 (0:00:02 remaining)

atharv@atharv:~$ nmap -A 10.10.10.6
Starting Nmap 7.80 ( https://nmap.org ) at 2025-03-26 09:52 IST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.21 seconds
atharv@atharv:~$ nmap -A 192.168.53.43
Starting Nmap 7.80 ( https://nmap.org ) at 2025-03-26 09:52 IST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.21 seconds
atharv@atharv:~$ nmap -A 192.168.52.43
Starting Nmap 7.80 ( https://nmap.org ) at 2025-03-26 09:52 IST
Stats: 0:01:02 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 91.67% done; ETC: 09:53 (0:00:02 remaining)

atharv@atharv:~$ sudo cat /var/log/snort/alert
cat: /var/log/snort/alert: No such file or directory
atharv@atharv:~$
```

Step 6: Simulate Port Scanning Attack

1. Run Hydra Brute-Force Attack on SSH:
hydra -L users.txt -P passwords.txt ssh://10.10.10.6

(Please create the user.txt and passwords.txt file by using mkdir command then Add some sample user as “test” and password as “toor” on the 10.10.10.6 system. Perform this attack from another VM like Kali Linux with ip as 10.10.10.4. Please remember all the IPs given may vary from system to system)

2. Check Snort Rules:
sudo nano /etc/snort/snort.conf
include $RULE_PATH/local.rules (Look for this line in the configuration file)
sudo nano /etc/snort/rules/local.rules (Edit this file)

alert tcp any any -> $HOME_NET any (msg:"Port scan detected"; flags:S; threshold: type threshold, track by_src, count 10, seconds 60; sid:1000001; rev:1;)

alert tcp any any -> $HOME_NET 22 (msg:"SSH brute force attack detected"; flow:to_server,established; detection_filter:track by_src, count 5, seconds 30;

sid:1000002; rev:1;)
(Add these 2 rules in the local.rules file)

3. Check Snort Logs:
sudo cat /var/log/snort/alert

```
[DATA] max 4 tasks per 1 server, overall 4 tasks, 9 login tries (l:3/p:3), ~3 tries per task
[DATA] attacking ssh://192.168.0.0:22/
[ERROR] could not connect to ssh://192.168.0.0:22 - Timeout connecting to 192.168.0.0
atharv@atharv:~/Documents/IS$ sudo nano /etc/snort/snort.conf
[sudo] password for atharv:
atharv@atharv:~/Documents/IS$ sudo nano /etc/snort/rules/local.rules
atharv@atharv:~/Documents/IS$ sudo nano /etc/snort/rules/local.rules
atharv@atharv:~/Documents/IS$ sudo cat /var/log/snort/alert
03/26-10:07:54.110227  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff15:e740
03/26-10:09:09.070067  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff9e:ac6b
03/26-10:09:17.721530  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff88:aaa
03/26-10:09:55.601856  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ffba:18be
03/26-10:09:56.137214  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:09:59.817696  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:09:59.823467  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:10:15.699575  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff5c:c4db
03/26-10:10:16.471064  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:10:18.674477  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff49:62bb
03/26-10:10:20.470464  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:10:20.474924  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:10:40.651393  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:10:41.122069  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:10:41.563762  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff9e:ac6b
03/26-10:10:41.769862  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:10:42.063731  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff9e:ac6b
03/26-10:11:01.428111  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff5a:275
03/26-10:11:06.074387  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff61:f1bf
03/26-10:11:15.319897  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:11:15.323919  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:11:17.174464  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ffeb:4253
03/26-10:11:23.072880  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:11:23.076822  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:11:28.348770  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:11:35.753860  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:11:36.122766  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff1c:746f
03/26-10:11:38.226077  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:11:39.135898  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::16
03/26-10:11:39.464188  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::16
03/26-10:11:39.572238  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:11:39.620188  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ffbc:2d8c
03/26-10:11:39.664016  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:11:49.265046  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:11:49.449937  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:11:49.670468  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff49:62bb
03/26-10:11:50.226882  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:11:50.678604  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff49:62bb
03/26-10:12:31.598159  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ffd7:f8e7
03/26-10:12:31.599387  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::16
03/26-10:12:31.816478  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::16
03/26-10:12:32.725350  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff5c:c4db
03/26-10:12:33.723483  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:13:26.297169  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:13:26.669833  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff32:ad4
03/26-10:17:42.584096  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff16:c89e
atharv@atharv:~/Documents/IS$
```

Part 4: Intrusion Analysis and Prevention

Step 7: Analyze IDS Logs

1. View Snort Alerts:

sudo tail -f /var/log/snort/alert

```
atharv@atharv:~/Documents/IS$ sudo tail -f /var/log/snort/alert
03/26-10:11:50.226882  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:11:50.678604  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff49:62bb
03/26-10:12:31.598159  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ffd7:f8e7
03/26-10:12:31.599387  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::16
03/26-10:12:31.816478  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::16
03/26-10:12:32.725350  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff5c:c4db
03/26-10:12:33.723483  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:13:26.297169  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:13:26.669833  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff32:ad4
03/26-10:17:42.584096  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff16:c89e
^C
atharv@atharv:~/Documents/IS$
```

Step 8: Mitigate Vulnerabilities

1. Block Attacker's IP Address:

sudo ufw deny from 10.10.10.4

2. Harden SSH Configuration:

sudo nano /etc/ssh/sshd_config (Please change the following in the file

PermitRootLogin no AllowUsers test

3. Restart SSH:

sudo systemctl restart ssh

```
atharv@atharv:~/Documents/IS$ sudo ufw deny from 192.168.0.0
Skipping adding existing rule
atharv@atharv:~/Documents/IS$ sudo nano /etc/ssh/sshd_config
atharv@atharv:~/Documents/IS$ sudo systemctl restart ssh
atharv@atharv:~/Documents/IS$
```

Experiment No. 6: Data Privacy and Encryption

Aim:
To protect sensitive data using encryption so that only authorized users can read it.

Step 1: Create a File with Sensitive Data

Linux:
```
touch secure_file.txt
echo "Top Secret Data" > secure_file.txt
```

Windows:
```
echo Top Secret Data > secure_file.txt
```

Explanation:
This step creates a plaintext file which contains readable data.

Step 2: Verify File Content

Linux:
```
cat secure_file.txt
```

Windows:
```
type secure_file.txt
```

Expected Output:
Top Secret Data

Step 3: Encrypt the File

Linux:
```
sudo apt update
sudo apt install openssl -y
openssl enc -aes-256-cbc -salt -in secure_file.txt -out secure_file.enc
```

Windows:
```
openssl enc -aes-256-cbc -salt -in secure_file.txt -out secure_file.enc
```

Explanation:
Encryption converts readable data into unreadable ciphertext using AES-256 algorithm.

Step 4: Verify Encrypted File

Linux:
```
cat secure_file.enc
```
Windows:
```
type secure_file.enc
```

Expected Output:
Random unreadable characters

Step 5: Decrypt the File

Linux:
```
openssl enc -aes-256-cbc -d -in secure_file.enc -out decrypted.txt
```

Windows:
```
openssl enc -aes-256-cbc -d -in secure_file.enc -out decrypted.txt
```

Explanation:
Decryption restores ciphertext back to original plaintext using the same key.

Step 6: Verify Decryption

Linux:
```
cat decrypted.txt
```
Windows:
```
type decrypted.txt
```
Expected Output:
Top Secret Data

Observations

- Plaintext file was readable initially
- After encryption, data became unreadable
- Decryption restored original data

Conclusion

Encryption ensures confidentiality by protecting sensitive data. Only authorized users with the correct key can access the information.

Experiment No. 7: Digital Forensics Basics

Aim:
To analyze system logs and understand how digital evidence is collected and examined for forensic investigation.

Step 1: Navigate to Log Directory

Linux:
cd /var/log

Windows:
Event Viewer can be opened using: eventvwr

Explanation:
This step accesses the location where system logs are stored.

Step 2: View Log Files

Linux:
ls

Windows:
Use Event Viewer to browse logs (System, Application, Security).

Explanation:
Lists all available log files which store system activity records.

Step 3: Monitor Logs in Real-Time

Linux:
sudo tail -f /var/log/syslog

Windows:
Use Event Viewer → Refresh logs

Explanation:
Displays live updates of system activity, useful for tracking events.

Step 4: Analyze Logs

Linux:
cat /var/log/syslog
Windows:
Check logs in Event Viewer

Explanation:
Helps identify login attempts, errors, and suspicious activities.

Step 5: Search Specific Events

Linux:
grep "error" /var/log/syslog

Windows:
Use filter option in Event Viewer

Explanation:
Filters logs to find specific keywords like errors or unauthorized access.

Observations

- System logs contain records of all activities
- Real-time monitoring shows live system events
- Logs help detect unauthorized access attempts

Conclusion

Digital forensics helps in analyzing system logs to detect suspicious activities and gather evidence. Logs play a crucial role in identifying security breaches and system behavior.

Experiment No. 8: Network and Program Security

Aim:
To analyze network security by scanning open ports and identifying active services using network scanning tools.

Step 1: Install Nmap Tool

Linux:
```
sudo apt update
sudo apt install nmap -y
```

Windows:
Download and install Nmap from official website.

Explanation:
Nmap is a network scanning tool used to discover hosts and services on a network.

Step 2: Perform Network Scan

Linux:
```
nmap localhost
```

Windows:
```
nmap localhost
```

Explanation:
This command scans the local system to identify open ports and services.

Step 3: Analyze Output

Example Output:
```
PORT        STATE SERVICE
80/tcp   open  http
```

Explanation:
Open ports indicate active services. Port 80 shows a web server running.

Step 4: Scan Specific Port

Linux:
nmap -p 80 localhost

Windows:
nmap -p 80 localhost

Explanation:
Scans a specific port to check if it is open or closed.

Step 5: Detect Service Version

Linux:
nmap -sV localhost

Windows:
nmap -sV localhost

Explanation:
Identifies the version of services running on open ports.

Observations

- Nmap detected open ports on the system
- Port 80 indicated an active HTTP service
- Open ports can be potential vulnerabilities

Conclusion

Network scanning helps identify open ports and services. Securing unnecessary open ports is essential to prevent unauthorized access and attacks.

Experiment No. 9: Operating System Security

Aim:
To implement file permission and access control mechanisms in an operating system to prevent unauthorized access.

Step 1: Create a File

Linux:
```
touch secure_file.txt
```

Windows:
```
echo. > secure_file.txt
```

Explanation:
This creates a new file for testing access control.

Step 2: Add Data to File

Linux:
```
echo "Top Secret Data" > secure_file.txt
```

Windows:
```
echo Top Secret Data > secure_file.txt
```

Explanation:
Adds sensitive data to the file.

Step 3: Set File Permissions

Linux:
```
chmod 600 secure_file.txt
```

Windows:
Right-click file → Properties → Security → Restrict access

Explanation:
Permission 600 allows only the owner to read and write.

Step 4: Verify Permissions

Linux:
```
ls -l secure_file.txt
```

Windows:
Check Security tab in file properties

Explanation:
Displays current file permissions.

Step 5: Create New User

Linux:
```
sudo adduser testuser
```

Windows:
```
net user testuser password /add
```

Explanation:
Creates a new user to simulate unauthorized access.

Step 6: Switch User and Test Access

Linux:
```
su testuser
cat secure_file.txt
```

Windows:
Run as different user and try opening file

Expected Output:
Permission denied

Explanation:
The new user cannot access the file due to restricted permissions.

Observations

- File permissions restrict access to owner only
- Unauthorized users cannot read the file
- Access control prevents data leakage

Conclusion

Operating system security mechanisms like file permissions help protect sensitive data from unauthorized access. Proper access control ensures system security and confidentiality.

Experiment No. 10: Database Security and Auditing

Aim:
To monitor system activities and track file access using auditing tools to detect unauthorized access.

Step 1: Start Auditing Service

Linux:
sudo systemctl start auditd
Windows:
Open Command Prompt (Admin):
net start eventlog

Explanation:
Starts auditing/logging service.

Step 2: Enable Auditing Service

Linux:
sudo systemctl enable auditd

Windows:
sc config eventlog start= auto

Explanation:
Ensures service starts automatically.

Step 3: Check Service Status

Linux:
sudo systemctl status auditd

Windows:
sc query eventlog

Explanation:
Verifies service is running.

Step 4: Add Audit Rule

Linux:
sudo auditctl -w /home/user/secure_file.txt -p rwxa -k security_audit

Windows:
Use Local Security Policy → Audit Policy → Enable Object Access auditing

Explanation:
Tracks file access activities.

Step 5: Perform File Activity

Linux:
cat secure_file.txt

Windows:
type secure_file.txt

Explanation:
Generate logs by accessing file.

Step 6: View Audit Logs

Linux:
sudo ausearch -k security_audit
Windows:
Open Event Viewer → Windows Logs → Security

Explanation:
Displays recorded events.

Step 7: Check System Logs

Linux:
sudo dmesg | tail

Windows:
Get-EventLog -LogName Security -Newest 10

Explanation:
Shows recent system-level logs.

Observations

- Audit logs recorded file access events
- System tracked user activities
- Logs help detect unauthorized access

Conclusion

Auditing tools in both Linux and Windows help monitor system activities, detect unauthorized access, and improve system security.